10 Kinds of People

CPU TIME LIMIT	MEMORY LIMIT	DIFFICULTY
2 seconds	1024 MB	medium (3)

Remember that joke about "10 kinds of people in the world"? In case you don't:

> *There are 10 kinds of people in the world:*
> - *those who don't understand binary,*
> - *those who think there are only binary and decimal,*
> - *and those who realize there are other bases.*

This joke is about numbers in different bases, or *radices*. The notation for a number represented by digits D in base B is written as D_B. For example, 10_2 is the binary number 10 (which equals 2 in decimal), and 10_3 is the number 3 in decimal represented in base 3.

The joke is funny because $10_2 = 2_{10}$, but $10_3 = 3_{10}$, and $10_4 = 4_{10} = 100_2$.

In this problem, we will dig a little deeper into that idea. You are given two numbers without specified bases. Your task is to determine whether there exist bases for each number such that their decimal values are equal.

To represent a wider range of numbers, we use the following digit scheme:

- digits `0–9` represent values 0–9,
- letters `a–z` represent values 10–35,
- letters `A–Z` represent values 36–61.

For example, $1Z_{62} = 62 + 61 = 123_{10}$.

We will consider bases from 2 up to 7500. Even though we may not have enough characters to represent all digits in these bases, we can still find bases that make two numbers equal.

For instance:

- The numbers `10` and `4000` are equal if `10` is in base 4000 and `4000` is in base 10, since both equal 4000 in decimal.
- Similarly, `10` and `4000` are equal if `10` is in base 500 and `4000` is in base 5, since both equal 500 in decimal.

Input

The first line contains an integer N, the number of pairs of numbers.

$$1 \leq N \leq 500$$

Each of the next N lines contains two numbers A and B. Each number consists of 1 to 5 characters chosen from `[0–9, a–z, A–Z]`.

Output

For each pair, if there exist bases a and b such that $A_a = B_b$, output three decimal numbers:

$$\text{decimal} \quad \text{value of } A_a, \quad a, \quad b$$

If there are multiple triples (A_a, a, b) that satisfy $A_a = B_b$, any one of them will be accepted. Otherwise, output:

$$\text{CANNOT MAKE EQUAL.}$$

Reminder: $2 \leq a, b \leq 7500$.

Sample Input 1

```
4
11 10
1Z 123
10 4000
1111 3333
```

Sample Output 1

```
3 2 3
123 62 10
500 500 5
CANNOT MAKE EQUAL
```

Sample Input 2

```
4
75 4GH
teH9x eR8LJ
231Tq jIhE7
NVWng Oiing
```

Sample Output 2

```
11275 1610 48
490386017 64 76
CANNOT MAKE EQUAL
CANNOT MAKE EQUAL
```